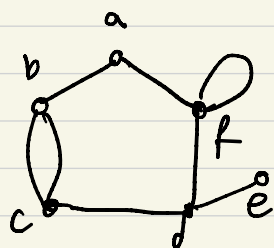
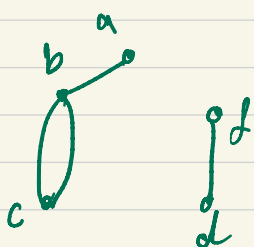


Subgraph: A graph $G' = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

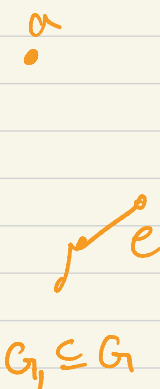


$G = (V, E)$

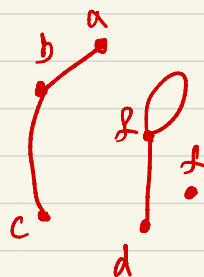


$G' = (V', E')$
 $V' = \{a, b, c, d\}$
 $E' = \{ \dots \}$

$$\Rightarrow G' \subseteq G$$



$G_1 \subseteq G$



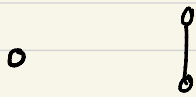
$G_2 \subseteq G$



spanning subgraph

If a subgraph contains all vertices of original graph then it is called a spanning subgraph.

Complete graph (K_n): a graph with n vertices s.t. there is an edge between any pairs of vertices



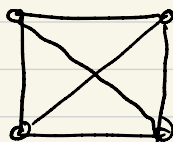
K_1



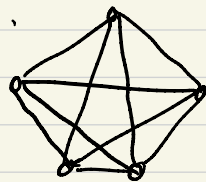
K_2



K_3



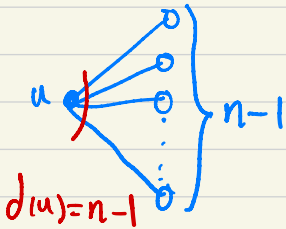
K_4



K_5

Q: What is the number of edges of K_n ?

First solution: Consider any vertex u of K_n



$$2|E| = \sum_{u \in K_n} d(u) = n \cdot (n-1)$$

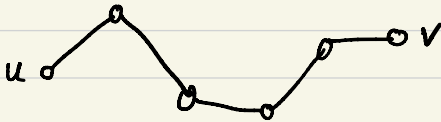
$$\Rightarrow |E| = \frac{n \cdot (n-1)}{2}$$

Second solution:

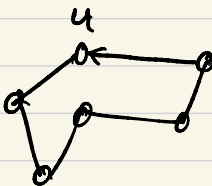
K_n is $(n-1)$ -regular

previous thm: $|E| = \frac{n \cdot d}{2} = \frac{n \cdot (n-1)}{2}$

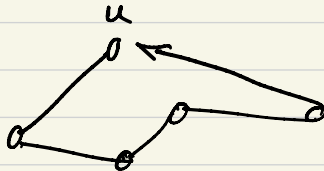
path: an alternating sequence of vertices and edges s.t. no vertex is repeated.



cycle: a path where the first and last vertices are the same.

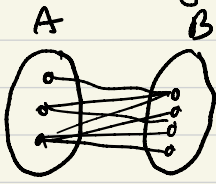


even-cycle
(even # of edges)

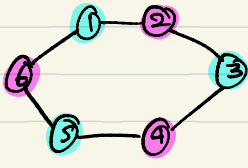


odd-cycle
(odd # of edges)

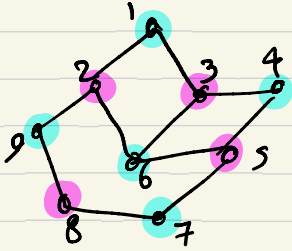
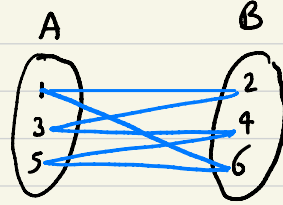
Bipartite Graph: The vertices can be partitioned into two sets A and B s.t. every edge has an endpoint in A and an endpoint in B .



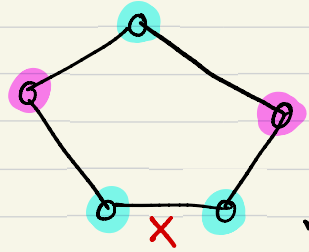
(no edge has both endpoints in A or in B)



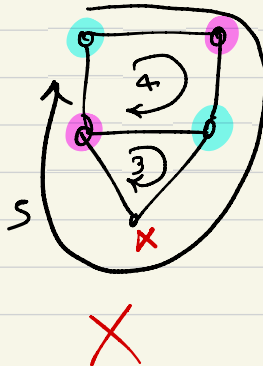
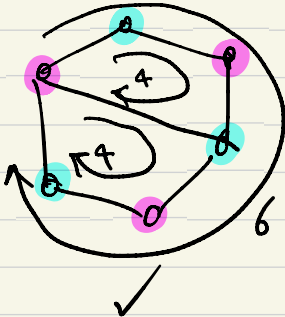
$A = \{1, 3, 5\}$
 $B = \{2, 4, 6\}$



$A = \{1, 4, 6, 9\}$
 $B = \{2, 3, 5, 7, 8\}$

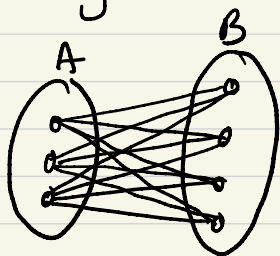


Theorem: A graph is bipartite if and only if it has no odd cycles.

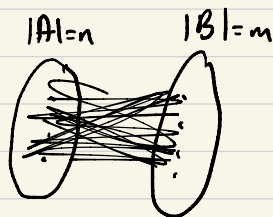


Complete Bipartite Graph ($K_{n,m}$) : $|A|=n$, $|B|=m$,

and every vertex of A is connected to every vertex of B .



$$K_{3,4} = K_{4,3}$$



$$K_{n,m}$$

Q.: How many edges do we have in $K_{n,m}$?

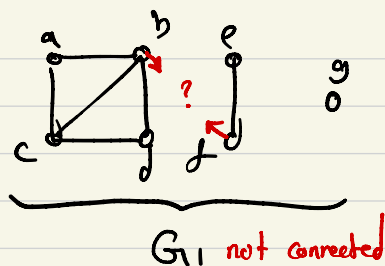
$$|A|=n, |B|=m$$

for every vertex in A we have m edges

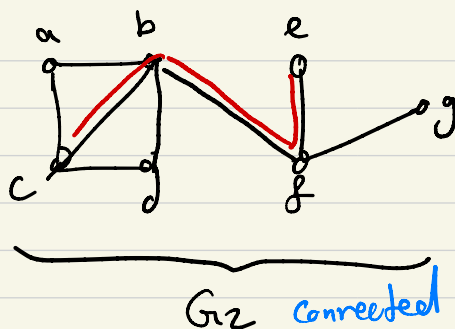
$\Rightarrow$ total number of edges is $n \cdot m$.

you can prove this by using sum of degrees.

G is connected if there is a path between any two vertices in G .

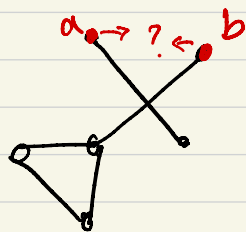


$$\omega(G_1) = 3$$

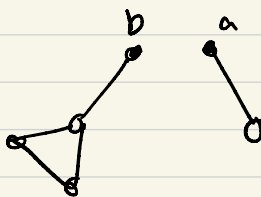


$$\omega(G_2) = 1$$

$\omega(G)$: $\neq 0$ of connected components of G .



$\cong$



$$\omega(G_3) = 2$$

Prove a statement for n (by induction)

- prove it for $n=0$ (base case)
- assume it is true for $n-1$ (hypothesis)
- prove it for n (ind. step)

} simple/weak induction

- prove for $n=0$

- assume it is true for any number less than n

- prove it for n

} strong induction

Theorem: Any graph G with n vertices and m edges has at least $n-m$ components. $\omega(G) \geq n-m$.



$$n=5$$

$$m=2$$

$$\omega(G) \geq n-m = 5-2 = 3$$



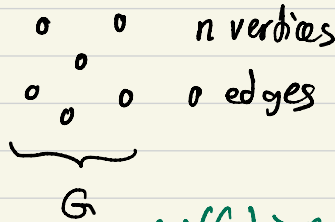
$$n=5$$

$$m=4$$

$$\omega(G) \geq n-m = 5-4 = 1$$

Proof (by induction): on # edges of graph, m

- Base case ($m=0$)



n vertices

0 edges

$$\omega(G) = n$$

G

$$\omega(G) \geq n-m = n-0 = n \quad \checkmark$$

- Induction hypothesis: assume the formula holds for any graph with n vertices and m' edges, where $m' < m$.

- Induction step: prove the formula for m edges.

G : n vertices, m edges

Remove an edge e from G and call the resulting graph G'

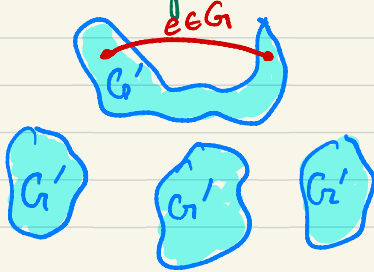
$$G' = G - \{e\}$$

$$\# \text{ edges of } G' = m' = m - 1$$

$$m' + 1 = m$$

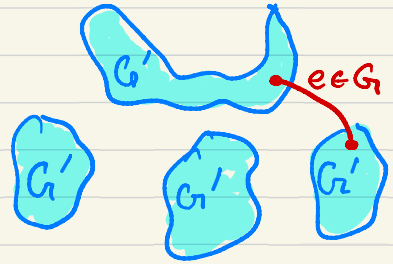
two case for e :

(a) both endpoints are in same component of G'



(a) $w(G) = w(G')$

(b) endpoints are in different components of G'



(b) $w(G) = w(G') - 1$

(a), (b) $\Rightarrow w(G) \geq w(G') - 1$

From induction hypothesis $w(G') \geq n - m'$
(*)

$$w(G) \geq w(G') - 1 \geq (n - m') - 1 = n - (m' + 1) = n - m \quad \square$$

Q: Prove that any connected graph with n vertices has $\geq n - 1$ edges.